

PRACTICAL COURSE WORKBOOK

Blockchain & Web3 Development

Build a clearer, auditable financial workflow

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a fictional financial model with assumptions and checks
SKILLS	Blockchain, Solidity, DeFi, Smart Contracts, Web3, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Blockchain appropriately in a realistic, bounded task.
- Use Solidity appropriately in a realistic, bounded task.
- Use DeFi appropriately in a realistic, bounded task.
- Use Smart Contracts appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners studying financial technology and analytical methods.

BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Blockchain & Web3 Development is a structured, practical course covering Blockchain, Solidity, DeFi, Smart Contracts, Web3. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Blockchain: Financial foundations	1. Value with Blockchain 2. Risk with Solidity 3. Markets with DeFi
02 Solidity: Tools and data	1. Records with Smart Contracts 2. Analysis with Web3 3. Controls with Blockchain
03 DeFi: Applied decisions	1. Scenarios with Solidity 2. Evaluation with DeFi 3. Reporting with Smart Contracts
04 Smart Contracts: Risk and governance	1. Security with Web3 2. Regulation with Blockchain 3. Review with Solidity

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Blockchain: Financial foundations

Apply Blockchain through value, risk, markets.

LESSON 01 / 18 MIN

Value with Blockchain

Learning objectives

- Explain value in the context of Blockchain & Web3 Development.
- Apply Blockchain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

value, Blockchain, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a value result produced with Blockchain?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Risk with Solidity

Learning objectives

- Explain risk in the context of Blockchain & Web3 Development.
- Apply Solidity to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

risk, Solidity, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a risk result produced with Solidity?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Markets with DeFi

Learning objectives

- Explain markets in the context of Blockchain & Web3 Development.
- Apply DeFi to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

markets, DeFi, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a markets result produced with DeFi?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Solidity: Tools and data

Apply Solidity through records, analysis, controls.

LESSON 04 / 30 MIN

Records with Smart Contracts

Learning objectives

- Explain records in the context of Blockchain & Web3 Development.
- Apply Smart Contracts to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

records, Smart Contracts, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a records result produced with Smart Contracts?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Analysis with Web3

Learning objectives

- Explain analysis in the context of Blockchain & Web3 Development.
- Apply Web3 to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

analysis, Web3, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a analysis result produced with Web3?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Controls with Blockchain

Learning objectives

- Explain controls in the context of Blockchain & Web3 Development.
- Apply Blockchain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

controls, Blockchain, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a controls result produced with Blockchain?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

DeFi: Applied decisions

Apply DeFi through scenarios, evaluation, reporting.

LESSON 07 / 26 MIN

Scenarios with Solidity

Learning objectives

- Explain scenarios in the context of Blockchain & Web3 Development.
- Apply Solidity to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

scenarios, Solidity, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a scenarios result produced with Solidity?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Evaluation with DeFi

Learning objectives

- Explain evaluation in the context of Blockchain & Web3 Development.
- Apply DeFi to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evaluation, DeFi, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a evaluation result produced with DeFi?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Reporting with Smart Contracts

Learning objectives

- Explain reporting in the context of Blockchain & Web3 Development.
- Apply Smart Contracts to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

reporting, Smart Contracts, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a reporting result produced with Smart Contracts?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Smart Contracts: Risk and governance

Apply Smart Contracts through security, regulation, review.

LESSON 10 / 22 MIN

Security with Web3

Learning objectives

- Explain security in the context of Blockchain & Web3 Development.
- Apply Web3 to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Web3, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a security result produced with Web3?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Regulation with Blockchain

Learning objectives

- Explain regulation in the context of Blockchain & Web3 Development.
- Apply Blockchain to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

regulation, Blockchain, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a regulation result produced with Blockchain?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Review with Solidity

Learning objectives

- Explain review in the context of Blockchain & Web3 Development.
- Apply Solidity to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

review, Solidity, finance

Retrieval prompt

In Blockchain & Web3 Development, which evidence best supports a review result produced with Solidity?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Blockchain & Web3 Development project using Blockchain, Solidity, DeFi.

DELIVERABLE	A working Blockchain & Web3 Development artefact plus an evidence-based self-review.
TOOLS	A suitable Blockchain environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Blockchain.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

National Payments System

Central Bank of Kenya - reviewed 2026-08-21

<https://www.centralbank.go.ke/national-payments-system/>

Financial education

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/financial-education.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Blockchain scope.